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Colorless Poly(amide-imide)s with Low Dielectric Constants and Enhanced Mechanical Properties through Tri-Armed Cross-Linkers

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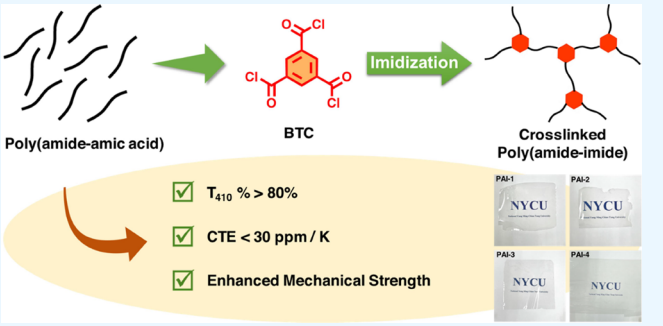
ABSTRACT: Colorless polyimides (PIs) have attracted a lot of attention because of the prosperous development of flexible devices. Traditional aromatic PIs possess great thermal stabilities and mechanical properties but lack high transparency. Aliphatic PIs provide great optical transparency, but the poor film formability, low mechanical strengths, and high coefficient of thermal expansion (CTE) make them hard to utilize practically. In this work, we present a series of colorless poly(amide-imide) films consisting of 4,4′-(hexafluoroisopropylidene)diphthalic anhydride (6FDA), trimetallic anhydride chloride (TMC), terephthalic chloride (TCL), and 2,2′-bis(trifluoromethyl)-[1,1′-biphenyl]-4,4′-diamine (TFMB) to overcome the coloration of aromatic PIs. In addition, the introduction of tri-armed cross-linker 1,3,5-benzenetricarbonyl trichloride (BTC) enhances the thermal properties and mechanical strengths. The resulting poly(amide-imide) films exhibit great thermal stabilities with the glass transition temperatures (T_g) of 342 to 351 °C and the CTE of 37 to 28 ppm/°C. In addition, attributed to the cross-linker BTC, these films possess excellent mechanical strengths from 90.1 to 130 MPa and low dielectric constants of 3.06 to 2.82. Most importantly, the cross-linked films still maintain high transparency. The optical properties of these films are marvelous in that the T_{410} is higher than 80% (82−86%) and the yellow indexes (YI) are lower than 2.5 (1.89−2.37). We believe these results can offer useful instructions for making colorless poly(amide-imide) films with a low CTE and high mechanical strength.



KEYWORDS: *colorless polymers, poly(amide-imide)s, high thermal stability, low dielectric constant, coefficient of thermal expansion, cross-linking reaction*

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■ INTRODUCTION

In recent years, due to the expeditious development of flexible electronic devices, lots of colorless materials have been taken into account and basically can be divided into two groups: thin glasses and polymers.[1] Among these two types of materials, polymers are more promising because of their low cost, lightweight as well as easy fabrication. Scientists have inquired about different kinds of polymers such as polyethylene terephthalate (PET), polyethylene naphthalate (PEN), poly(ether sulfone) (PES), and polycarbonate (PC).[2−7] All of these polymers provide low coloration as well as high transparency; however, the inferior thermal stabilities are their Achilles' heel. For organic light emitting diode (OLED) applications, for example, most polymers have coefficient of thermal expansion (CTE) values higher than 60 ppm/°C, leading to great difficulties in aligning OLED materials in layers; also, none of them possess glass transition temperatures (T_gs) higher than 250 °C, which are lower than the fabrication window of OLEDs.[8−10]

Polyimides (PIs) are considered high-performance polymers with the capabilities to reach T_g over 300 °C and low CTE. Their extraordinary thermal stabilities, outstanding mechanical strength, and relatively low dielectric constant have caught scientists' eyes.[11,12] Kapton, which is commercialized in the 1960s, given its well-performed thermal resistance properties, has been widely used in aerospace, electronic insulators, and flexible printed circuit boards (FPCBs).[13−16] The traditional fully aromatic PI films, however, display strong coloration and are deficient in optical transparency. The coloration stems from the inter- and intramolecular charge transfer complexes inside the highly conjugated molecular structures.[2,17,18] With the purpose of overcoming the coloration of PIs, several strategies are considered, such as introducing aliphatic, ether, and sulfone moieties into PI chains or designing monomers with bulky groups to reduce the stacking density.[18−21] All of these methods are willing to alleviate interchain charge transfer. Notwithstanding the decrease in coloration, the

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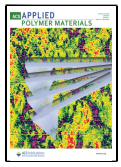

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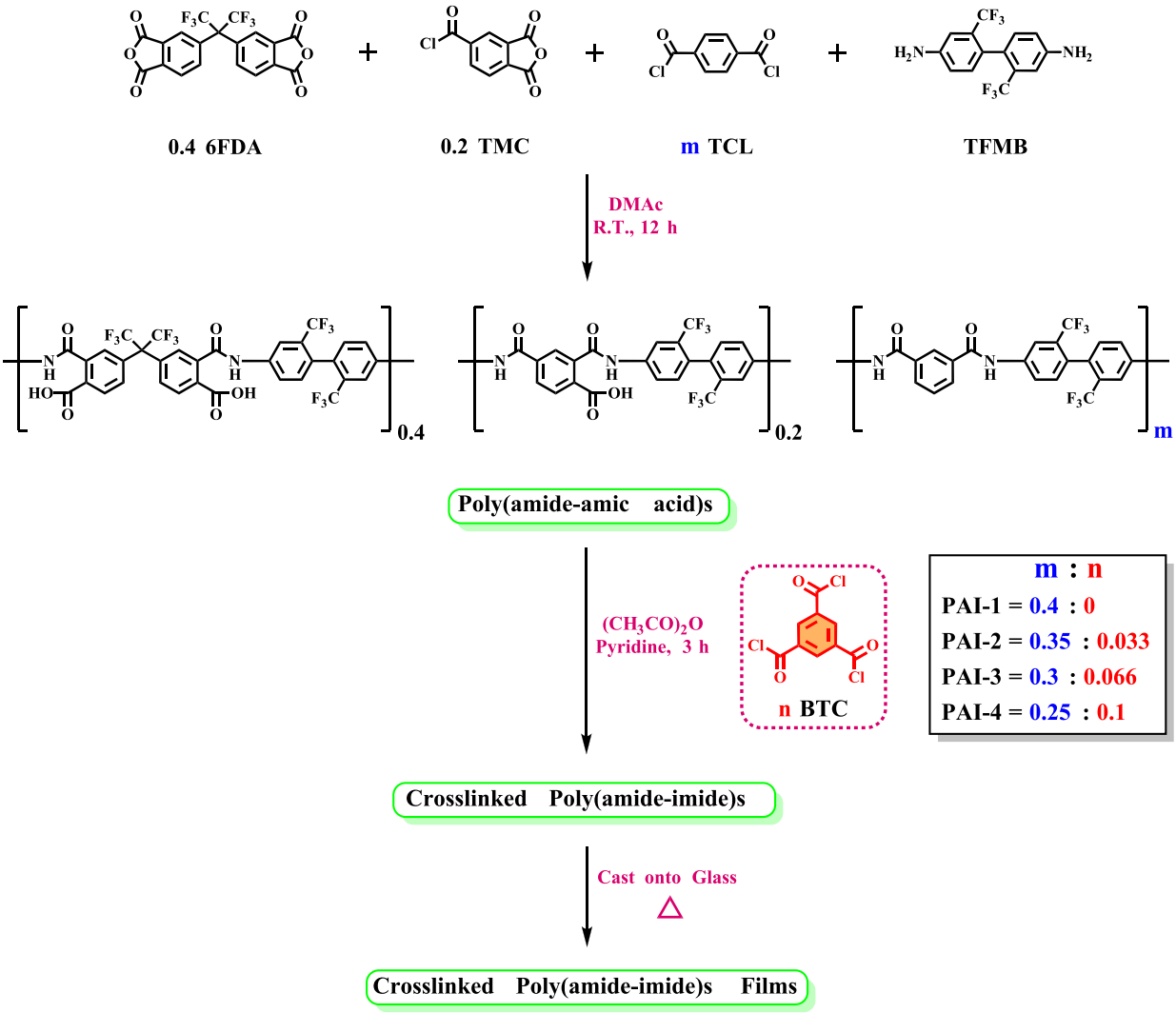
Scheme 1. Synthetic Scheme of the Colorless PAI Films



weak mechanical strength caused by the introduction of aliphatic groups and high CTE values arising from bulky groups have restricted their applications in optical fields.

To resolve these issues mentioned above, in this study, we adopt two main strategies. First, to decrease the CTE and the coloration of PIs, we suggest that poly(amide-imide) (PAI) is a good candidate. Previously, Fang et al. reported a PAI system of trimellitic anhydride chloride (TMC) with the cooperation of terephthalic chloride (TCL) and 2,2′-bis(trifluoromethyl)-[1,1′-biphenyl]-4,4′-diamine (TFMB).[22] These PAI films revealed relatively good optical transparency (T_{450} = 84.0%) as well as a low CTE (21 ppm/°C). Furthermore, Kim et al. synthesized a series of transparent poly(amide-imide)s utilizing diacid monomers with imide groups and diamines through amide formation reactions. As a result, the liquid crystalline-like polymer chains were beneficial for achieving ultralow CTE (<10 ppm/°C). Additionally, the film formation at a relatively low temperature can offer the utmost transparency (T_{550} = 88%).[23] Also, several reports have mentioned that the PI films formed by the 4,4′-(hexafluoroisopropylidene)diphthalic anhydride (6FDA) and TFMB exhibit extraordinary transparency, but the lack of low CTE values makes it hard to utilize.[24−27] Therefore, in this work, we chose 6FDA/TMC/TCL/TFMB to synthesize our PAIs, anticipating complementary effects from each component to achieve great transparency as well as low CTE.

Second, to improve the mechanical strength of PAI films, we adopt the strategy of cross-linking between PAI chains. Previously, Wu and co-workers demonstrated a series of PAI films using a trifunctional amine monomer 1,3,5-tri(4-aminophenoxy) benzene (TAPOB) as a cross-linker.[24] The improvement in mechanical strength and reduction in CTE are impressive. In addition, the microbranched structures in the polymer chains derived from TAPOB lead to a significant drop in dielectric constants, also demonstrating the reduction of the dielectric constants using multifunctional monomers. Moreover, Meador et al. reported a series of PI aerogels using 1,3,5-benzenetricarbonyl trichloride (BTC) as a cross-linker.[28] The modulus after cross-linking by BTC is comparable to or better than those using different cross-linkers. Considering the commercial availability and similar structure to TCL, in this work, we utilize BTC to prepare cross-linked colorless PAIs.

By adjusting the ratios between TCL and BTC, we synthesize a series of cross-linked colorless PAI films by chemical imidization followed by thermal imidization; the molar ratio of TCL varies from 25 to 40%, while BTC varies from 0 to 10%. The PAI films possess enhanced thermal properties with the 5% weight loss decomposition temperature

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($T_{d5\%}$) at around 490 °C, the T_g increased from 342 to 351 °C, and the CTE decreased from 37 to 28 ppm/°C. The mechanical strengths of the PAI films are also improved from 90.1 to 130.3 MPa. Impressively, the optical properties of the PAI films are within the range of colorless films after cross-linking; the T_{410} of these films is higher than 80% (82−86%), and the yellow index (YI) is lower than 2.5 (1.89−2.37). The dielectric constants of the PAI films also drop from 3.06 to 2.82 because of the three-dimensional cyclic structures by the A2+B3 polymerization. This study provides an effective and versatile strategy to prepare colorless polymer films for optoelectronic applications.

■ EXPERIMENTAL SECTION

Materials. In this work, the used materials are as follows. 4,4′(Hexafluoroisopropylidene)diphthalic anhydride (6FDA) was received from Jin-Yu Technology with 99%+ purity. 2,2′-Bis-(trifluoromethyl)benzidine (TFMB) was received from Shifeng Technology with 99%+ purity. Terephthaloyl chloride (TCL) was purchased from Acros. Trimellitic anhydride chloride (TMC) was purchased from Nova Materials. 1,3,5-Benzenetricarboxylic acid chloride (BTC) was purchased from Sigma-Aldrich. Methanol (CH_3OH) was received from J. T. Baker. *N*,*N*-Dimethylacetamide (DMAc) and dimethylformamide (DMF) were obtained from Acros Organics and stored under a nitrogen (N_2) atmosphere with 4 Å molecular sieves. All of these chemicals and solvents were used without any further purifications for constructing the PAI films.

Measurements. Inherent viscosities were measured in DMAc at a concentration of 0.5 g/dL at 30 °C by using an Ubbelohde viscometer. The ultraviolet−visible (UV−vis) spectra of all of the colorless PAI films (film thickness = 27.5 ± 2.5 μm) were acquired utilizing a UV−vis spectrophotometer (U-4100 Spectrometer, Hitachi) in the wavelength (λ) range of 300−1000 nm. The yellow index and haze values of the PAI films were measured by a Konica Minolta CM-5 colorimeter according to the CIE1976LAB standard. Attenuated total reflectance Fourier transform infrared (ATR-FT-IR) spectroscopy data of the PAI films were obtained from a Spectrum One (PerkinElmer) instrument. The mechanical properties of the PAI films were conducted using a Shimadzu EZ-L instrument in accordance with the ASTM D882 guideline. The films [100 mm (L) × 10 mm (W) × 27.5 ± 2.5 μm (T)] were stretched at a drawing rate of 12.5 mm min^{-1}. The measurements of CTE were tested at a heating rate of 5 °C min^{-1} by a TA Instruments TMA450EM, and the CTE values were calculated from 100 to 200 °C in the second heating cycle. The decomposition temperature ($T_{d5\%}$) data were collected by a TA Instruments TGA55 at a rate of 20 °C min^{-1} up to 700 °C. The glass transition temperatures (T_gs) were measured via a TA Instruments DMA850 at a rate of 3 °C min^{-1} up to 400 °C and with a load frequency (sinusoidal) of 1.0 Hz in the air, and the T_g values were calculated from peaks of the tan δ (E'/E'') curves. The dielectric constants and dissipation factors were determined by a high-frequency vector network analyzer Anritsu MS46122B with a kit of 10 GHz cavity D2014-1; the operating condition was managed at 22 °C and 50% humidity. The static contact angles of the PAI films with water were examined by a contact angle analyzer (First Ten Ångstrom, FTA125). The scanning electron microscopy (SEM) graphs were captured by a field emission scanning electron microscope (JEOL, JSM-7401F). The weight-averaged molecular weights (M_w), number-averaged molecular weights (M_n), and polydispersity index (PDI) of all the PAI powders were acquired from a Jasco gel permeation chromatography (GPC) system. Polystyrenes were used as the standard. DMF was used as the mobile phase of chromatography. The flow rate and temperature were set to 0.5 mL min^{-1} and 50 °C, respectively. A water absorption test was conducted on the PAI films. In this process, all of the films were immersed in deionized water at 25 °C, and the weight differences of all of the films were measured after 24 h. Fractional free volume (FFV) was calculated as follows: FFV = $(V - V_0)/V$, $V_0 = 1.3\ V_w$, V is the molar volume of the repeating unit (cm^3 mol^{-1}), V_0 is the volume occupied by the molecular chain (cm^3 mol^{-1}), and V_w is the van der Waals volume. V is calculated from the formula: d (density) = M/V. The densities of the films were measured via a Mettler−Toledo balance ME204T coupled to a density kit utilizing Archimedes' principle. V_w is calculated via the group contribution Bondi method.[29]

Syntheses of the Poly(amide-amic acid) Solutions. In this work, TFMB/6FDA/TMC were used in the same molar ratio, while the ratios of TCL and BTC were denoted in Scheme 1. Take PAI-2 as an example: TFMB 3.016 g (9.418 mmol) and DMAc (40 mL) solvent were added into a three-necked flask fitted with a mechanical stirrer under a N_2 atmosphere. After TFMB was completely dissolved, 6FDA 1.673 g (3.767 mmol) was then added to the solution and agitated for 3 h. TCL 0.669 g (3.297 mmol), TMC 0.396 g (1.884 mmol), and pyridine 0.46 mL (5.711 mmol) were added to the previous solution and agitated for an additional 12 h.

Preparation of the Cross-Linked PAI Powders. Chemical imidization was performed one hour before adding the cross-linker BTC. Take PAI-2 as an example: acetic anhydride 2.13 mL (22.53 mmol) and pyridine 1.82 mL (22.59 mmol) were added to the previous solution. Agitated at room temperature for an hour, the cross-linker BTC 0.083 g (0.313 mmol) was diluted by DMAc (30 mL) and then added to the solution drop by drop to prevent gelation. The overall wt % of the solution was ca. 8%. After the addition of BTC, the solution was heated to 80 °C for 2 h. After the sequence, the solution was slowly poured into 1 L of methanol. The PAI powder was collected after filtration and dried at 130 °C for 12 h. The yield of PAI powder was 83% (4.45 g).

Preparation of the PAI Films. All of the PAI films in this work were prepared by the same method. The prepared PAI powders were dissolved in DMAc at 15 wt % to obtain the PAI solutions. The solutions were cast onto a dust-free glass plate and scraped by an automatic coating machine with controlled thicknesses. Then, the films were sent into a preheated muffle in a sequence of 70 °C for 1 h, 150 °C for 1 h, 220 °C for 2 h, followed by cooling to R.T. Finally, the PAIs/glass were soaked into deionized water to release the PAI films from the substrates.

■ RESULTS AND DISCUSSION

Synthesis and Characterizations of the PAI Films. As shown in Scheme 1, PAIs are synthesized via a two-step polymerization method. After polymerizing for 12 h and chemical imidization, the cross-linker BTC is added to the solution at a dilute condition to prevent gelation. By chemical imidization, the thermal imidization temperature can be utilized down to 220 °C to alleviate the coloration during the thermal imidization process. The chemical structures of all the PAI films are measured by attenuated total reflectance Fourier transform infrared spectrometer (ATR-FT-IR). All of the spectra of the PAI films are shown in Figure 1. The characteristic peaks at 1722 and 1787 cm^{-1} denote the symmetry and asymmetry peaks of imide C═O, respectively; the C−N stretching of the imide peak appears at 1367 cm^{-1}. The peak at 1684 cm^{-1} shows the amide symmetry C═O. The peaks at 1306, 1252, and 1169 cm^{-1} represent that the CF_3 groups appear inside the 6FDA and TFMB moieties.[30] However, there is no significant difference between these films because the same ratios of the amide and imide groups are maintained.

Optical Properties of the PAI Films. The optical properties of the PAI films are determined by a UV−vis spectrometer and colorimeter. The UV−vis graph and real film picture of the PAI films are illustrated in Figure 2. The yellow index (YI), haze, and detailed optical properties of the PAI films are shown in Table 1. The transmittances of the PAI films are enormously affected by the charge transfer between the

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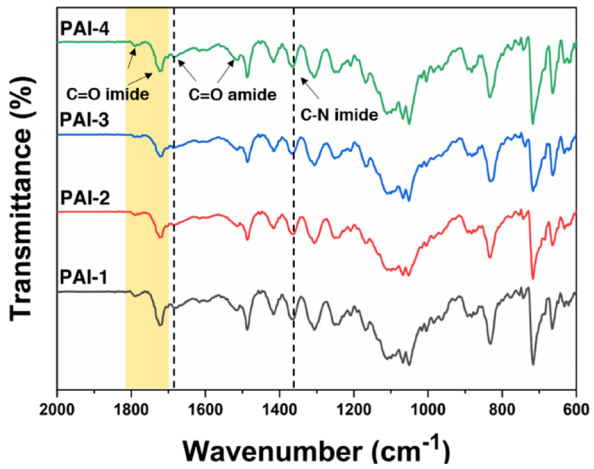

Figure 1. ATR-FT-IR spectra of the PAI films.

molecular chains. In this work, the introduction of the trifluoromethyl groups inside the monomer 6FDA and TFMB can not only increase the free volume but also reduce the intermolecular charge transfer.[26,31,32] Also, the cross-linker BTC consists of three acyl chloride groups. Therefore, the amide bonds are formed after the reaction with diamine TFMB, which maintains the transparency of the PAI films. Compared with the other PAI films, PAI-1 shows the best transmittance at 410 nm (T_{410} = 86%). The absence of the cross-linker BTC indicates that the distances between different polymer chains are not restricted; thus, the charge transfer between the chains is less than those with cross-linkers. From PAI-2 to PAI-4, the T_{410} values marginally drop down to 82% because of the increased content of BTC, leading to local chain aggregation near the cross-linking site.[18] Furthermore, from the SEM images (Figure S1), some voids are observed in the cross-linked films. However, some articles on PI/silica composite films reported that the transmittance is decreased with lots of voids and we regard this phenomenon as the phase separation effect between the PI substrate and silica nanoparticles.[33] Even though the content of the cross-linker can lead to a slight decrease in transmittance, still it can remain within the range of the transparent films.

When it comes to colorless PI/PAI, the yellow index (YI) of the PI films is a significant indicator of the coloration of the films. The results are summarized in Table 1. In our work, we utilize a colorimeter to assess the coloration of the PAI films in accordance with the CIE1976Lab of the D65/10° system. The L^* value represents the lightness of the film, while 0 indicates black and 100 indicates white. All of the PAI films in this work are in the range of 95.92−96.00, demonstrating that all of the films possess excellent brightness. The a^* and b^* values denote the extent of the red/green and yellow/blue colors of the films. As both the a^* and b^* values are closer to 0, the colors of the films are lighter. From PAI-1 to PAI-4, it is observed that both the a^* and b^* values are slightly leaving from the origin point, representing the light coloration of the PAI films after cross-linking. Furthermore, the YI values of PAI-1 to PAI-4 are in the range of 1.89−2.37, manifesting that all the PAI films in this work are given light colors and can be seen as colorless films.

Thermal Stabilities and Mechanical Properties of the PAI Films. The thermal characteristics of the PAI films are measured by a thermogravimetric analyzer (TGA). The TGA curves are displayed in Figure 3a. The T_d is basically related to the bond energy and constituent group inside the molecular chain. Thanks to the $-CF_3$ moiety inside the dianhydride 6FDA and diamine TFMB, the bonding energy of C−F is as strong as benzene moiety (C−F = 532 kJ/mol, C═C benzene = 518 kJ/mol).[34] As a result, the PAI films in this work have $T_{d5\%}$ around 490 °C. Also, the amide bond ratios of all the films are controlled at 50%, and the remaining part of the imide bond is held at 50% as well; hence, no significant difference of the $T_{d5\%}$ is observed.

The T_gs of all PAI films are tested via a dynamic mechanical analyzer. The T_gs are calculated via the onset point of tan δ curves, as shown in Figure 3b. Basically, T_g is correlated to the rigidity and linearity of the molecular chain. The more rigid the chain is, the more energy it needs to move the chain.[35] In this work, the multiarmed cross-linker BTC is capable of pulling different chains closer, thus tightening the chain packing. From PAI-1 to PAI-4, the T_g increases from 342 to 351 °C; a proliferation in T_g indicates the effort of BTC to link chains together, which restricts the motions of the polymer chains.

Thermal expansion is considered a critical property in colorless PAI because of the prevention of rift arising from the CTE difference between the substrate and metal lines. The CTE values in this work are analyzed in the range of 100−200 °C, as shown in Figure 3c. Typically, the CTE value is highly related to the linearity of the polymer chain. The TCL and TMC moieties in this PAI series provide linearity from the

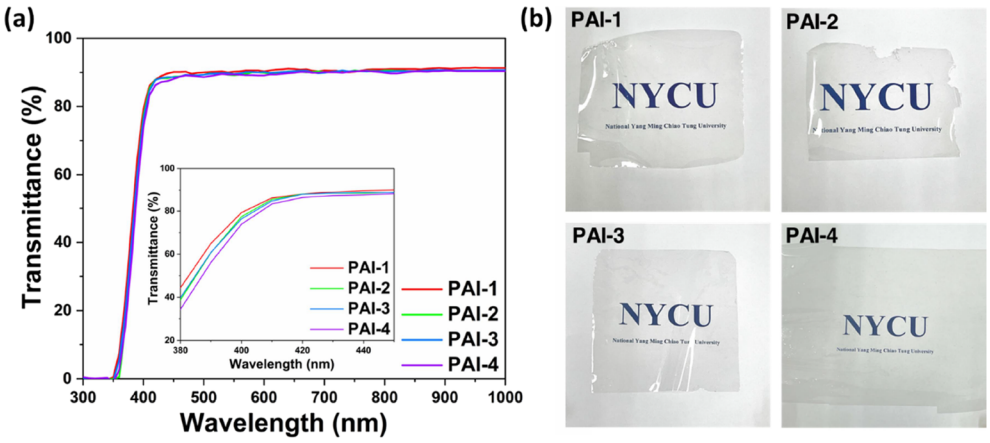

Figure 2. (a) UV−vis spectra of the PAI films. (b) Real photographs of the PAI films.

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Table 1. Optical Properties of the PAI Films

PAI films	λ_{cutoff} (nm)	T_{410}[a] (%)	T_{total}[b] (%)	L^*	a^*	b^*	YI[c]	haze
PAI-1	349	86.0	89.8	95.92	−0.21	1.10	1.89	0.76
PAI-2	351	85.0	89.2	96.00	−0.25	1.17	1.93	1.17
PAI-3	354	83.1	88.7	95.94	−0.41	1.37	2.37	0.86
PAI-4	354	82.2	88.6	95.96	−0.35	1.49	2.34	1.15

[a]Transmittance at a wavelength of 410 nm. [b]Average transmittance in the range of 400−760 nm. [c]Yellow index.

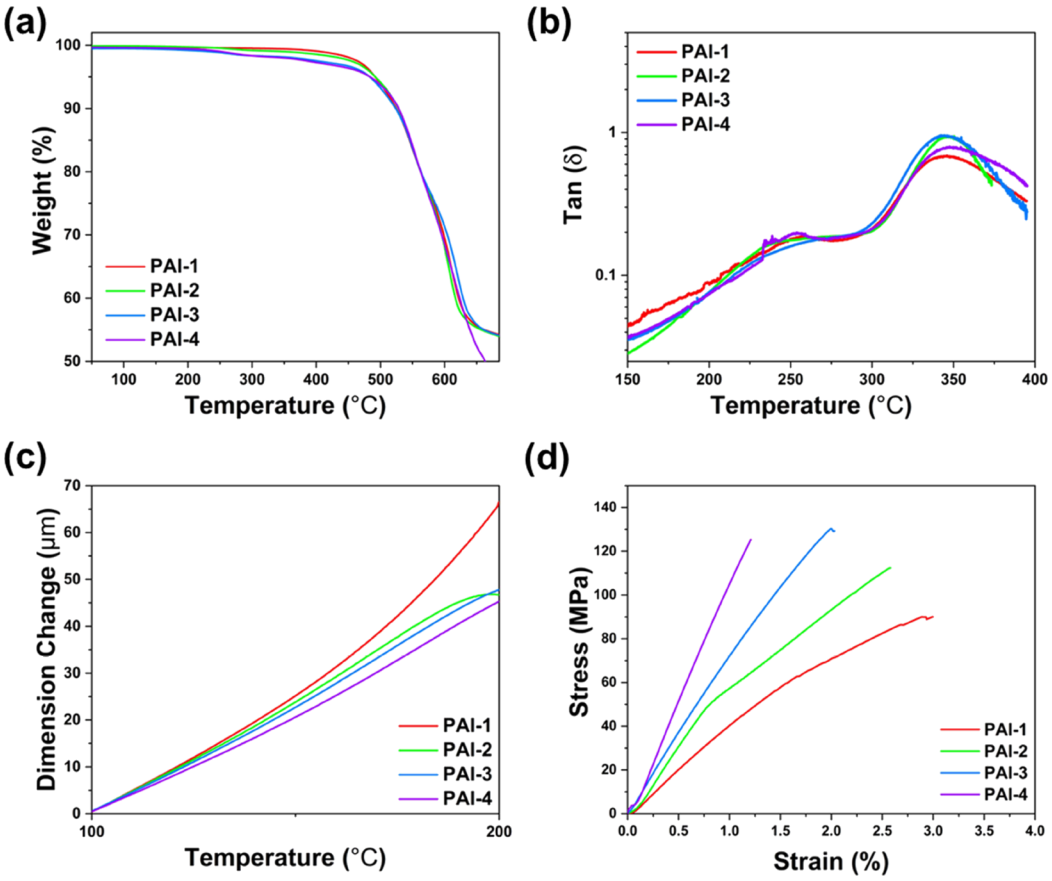

Figure 3. (a) TGA, (b) DMA, and (c) TMA thermograms and (d) stress−strain curves of the PAI films.

Table 2. Thermal, Optical, and Mechanical Properties and Inherent Viscosity of the PAI Films

PAI films	T_g (°C)[a]	$T_{d5\%}$ (°C)[b]	CTE (ppm/°C)[c]	σ (MPa)[d]	E (GPa)[e]	ε (%)[f]	η_{inh}[g] (dL/g)
PAI-1	342	496	37.0	90.1	2.7	3.02	0.63
PAI-2	347	494	31.5	112.4	4.4	2.58	0.96
PAI-3	350	491	30.3	130.3	6.5	2.03	1.06
PAI-4	351	490	28.4	125.3	9.5	1.15	1.24

[a]Glass transition temperature. [b]Decomposition temperature at 5% weight loss. [c]Coefficient of thermal expansion. [d]Tensile strength. [e]Initial modulus. [f]Elongation at break. [g]Inherent viscosity.

benzyl group. Also, the $-CF_3$ groups inside diamine TFMB can restrict the rotation of the single bond.[36] Therefore, from the result of PAI-1, we obtain the CTE value down to 37.0 ppm/°C. Moreover, the cross-linker BTC links different main chains together, thus restricting the rotation or twist of the polymer chains during the heating program of the CTE tests.[24,37] From the results of PAI-2 to PAI-4, the CTE values range from 31.5 to 28.5 ppm/°C, a significant decrease while comparing with the uncross-linked PAI-1, manifesting great potential in the display field.

The moduli, tensile strength, and elongation at the break of the colorless PAIs are summarized in Table 2 and Figure 3d. The inherent viscosity (η_{inh}) values and molecular weights (M.W.) of PAI-1 to PAI-4 are displayed in Tables 2 and S1, respectively. Compared with the TMC-TFMB system, PAI-1 in this work shows worse mechanical strength and elongation at break owing to the presence of the 6FDA moiety that leads to loosened molecular chains.[38,39] PAI-1 has the lowest inherent viscosity of 0.63 and the lowest molecular weights. After cross-linking, the inherent viscosities of PAI-2 to PAI-4 increase from 0.96 to 1.24; also, the M_n has a significant surge from 9.4×10^4 to 1.3×10^5 (g/mol). Not only the M_n values of the cross-linked PAIs are increased, but also the polymer chain is tensed up after cross-linking.[24,37] Hence, the mechanical

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strength can improve from 90.1 MPa (PAI-1) to 130.3 MPa (PAI-3). Despite some voids observed in the SEM images, as displayed in Figure S1, we deem that the possibility of the cracks caused by the voids is overwhelmed by the power of cross-linking.[40−42] As for the cross-linked PAI systems, the decrease in elongation is unpreventable.[37,43,44] The polymer chains are tightened together and form netlike microstructures leading to worse mobility of polymer chains. Therefore, from our results, the elongation degrees of the films decreased from 3.02 to 1.15%, which can be seen as an indicator of successful cross-linking.

Dielectric Properties of the PAI Films. Dielectric properties are considered one of the most appealing assets in PIs. In general, there are several strategies to achieve low dielectric constants in designing novel monomers, such as increasing the FFV of the PAI molecule by adding some nonplanar or bulky groups or reducing the polarity via introducing fully aliphatic components.[45−47] In this work, the introduction of the cross-linker BTC is also advantageous for lowering the dielectric constants. The values of the dielectric constants and dissipation factors are described in Figure 4, and the detailed values of density and FFVs are summarized in Table 3.

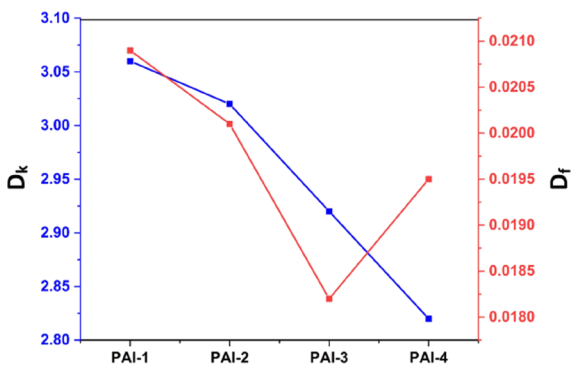

Figure 4. Dielectric constant (D_k) and dissipation factor (D_f) values of the PAI films.

It is observed that the dielectric constants shrink from 3.06 to 2.82 (PAI-1 to PAI-4); we deem that this result stems from the introduction of the cross-linker that leads to the mini pores inside the films. Additionally, it is well-known that an A2 + B3 polymerization could lead to hyperbranched or three-dimensional cyclic structures.[48] Hence, SEM is utilized to verify the results. The cross-sectional SEM images of the films are shown in Figure S1. From the SEM images, we observed significant differences between PAI-1 and PAI-4. PAI-1 displays a smooth surface while PAI-4 has a lot of pores and oval-like or irregular structures, expressing the ability of cross-linking to decrease the dielectric constants.

Several indirect proofs allow us to confirm the cross-linking via BTC, such as the FFV and density results. The FFV and density results of the PAI films are shown in Table 3. From PAI-1 to PAI-4, the densities of the films decreased from 1.587 to 1.313 g/cm^3, indicating that the tiny pores created inside are filled with air. FFV is an important index in the gas permeation field. Recently, researchers adopted this concept for dielectric properties. Higher FFV represents that the aggregation of polymer chains are loosened, thus leading to better dielectric constants. The FFV values increase from 0.148 to 0.190, which also indicates that the films are filled with air.

Furthermore, the degrees of water uptake of the films are examined by water contact angles and water absorption tests, and the results are shown in Figure S2 and Table 3. Zhang and co-workers demonstrated evidence that the large dipole of water has caused serious effects on dielectric constants; the composites with better hydrophobicity are able to reach lower dielectric constants.[49] In this work, the water contact angles increase from 64 to 79° as the content of the cross-linker BTC increases from 0 to 10%. Additionally, the decreasing water uptakes from 1.48 to 1.03% further verify the improved hydrophobicity of the PAI films, which shows improvements while comparing with previously reported PI and PAI.[50,51] These results indicate that the microbranched structures are helpful to enhance the hydrophobicity of the films, thus resulting in the improvement of the dielectric constants.

CONCLUSIONS

In this work, we present a series of cross-linked colorless PAI films. All of these films are prepared by chemical imidization followed by thermal imidization. From the thermal analysis results, the $T_{d5\%}$ of these films are all observed at around 490 °C because the amide and imide groups possess the same ratios. The T_g increases from 342 to 351 °C, the CTE decreases from 37 to 28 ppm/°C, and the mechanical strength enhances from 90.1 to 130.3 MPa. These results manifest that the polymer chains are pulled closer and tightened after adding the cross-linker BTC. Remarkably, the optical properties are within the range of colorless films after cross-linking; the T_{410} of these films is higher than 80% (82−86%), and the yellow index (YI) is lower than 2.5 (1.89−2.37). Because the A2+B3 polymerization can form three-dimensional cyclic structures, the dielectric constants drop from 3.06 to 2.82, and evidence is provided by SEM. All in all, we deem that the cross-linker BTC is effective in enhancing the dielectric and mechanical properties without compromising the great transparency of the films, wishing to provide useful knowledge in cross-linked colorless PI fields.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsapm.3c02287.

Table 3. Dielectric Properties, Density, and FFV of the PAI Films

PAI films	D_k[a]	D_f[b]	V_{molar}[c]	density	FFV[d]	contact angle (deg)	water absorption (%)
PAI-1	3.06	2.09×10^{-2}	391.4	1.587	0.148	64	1.48
PAI-2	3.02	2.01×10^{-2}	423.5	1.474	0.167	67	1.17
PAI-3	2.92	1.82×10^{-2}	446.2	1.406	0.169	73	1.06
PAI-4	2.82	1.95×10^{-2}	480.4	1.313	0.190	79	1.03

[a]Dielectric constant. [b]Dissipation factor. [c]Molar volume. [d]Fractional free volume.

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SEM images and water contact angle and GPC data are included in the Supporting Information (PDF)

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Notes

The authors declare no competing financial interest.

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